ZKSF
Zero Kelvin Simulation Foundry

ZHF-v0.1
2026-10-28T11:20:53+00:00

Certificate of Hardware Fidelity

Certificate ID
3b7a1267e203441a

Measured hardware fidelity: 0.2525 vs. exact ideal

ENGINE
qpu.rigetti
SHOTS
1000
HELLINGER FIDELITY
0.2525

METHOD
quantum hardware (Rigetti Cepheus-1-108Q)
DEVICE
Cepheus-1-108Q
VERIFICATION MODE
direct

MEASURED OUTCOMES (top)

011	208 shots
111	180 shots
001	147 shots
101	111 shots
010	110 shots
110	101 shots

HOW THIS CERTIFICATE WAS PRODUCED

• This certificate reports ZHF-v0.1, the hardware fidelity protocol: how closely a real quantum processor's measured output matches the ideal distribution for the same circuit, as opposed to the accuracy of a classical approximation.

• Verification mode is direct: the same circuit was additionally simulated exactly on a classical computer (feasible here because it uses at most 24 qubits). The reported Hellinger fidelity is the statistical overlap between the hardware's measured counts and that exact ideal distribution, on a scale from 0 (no overlap) to 1 (identical).

• Counts are raw hardware output: no error mitigation, post-selection, or readout correction has been applied. Any deviation from the ideal distribution reflects the physical noise of the device.

Independently verifiable at:
https://api.zksf.org/certify/3b7a1267e203441a

Circuit SHA-256 99b41c00d8345c3a2075bce453e75bcb43a1a6af0dc51855bf759b4482643fee